

ANALYSTLAB AFRICA EXPERIENCE LAB

Official Week 6 Project Summary — Machine Learning Engineering Track

Intern Name: Grai Rudolf

Track: Machine Learning Engineering

Project Title: HealthConnect Clinic Appointment Attendance Optimization

Date: September 2026

1. What I Planned to Accomplish

Build on the Week 5 modular pipeline by integrating validation checks, structured logging, improved (tuned) model configurations, an end-to-end pipeline class, and integration testing. Collaborate with the Data Science track to integrate their candidate model into the pipeline.

2. What I Completed

1. Added `validation.py` with 5 validation checks covering raw data, clean data, features, preprocessor, and prediction output.
2. Created `model_integration.py` with the `HealthConnectPipeline` class wrapping the full load-to-predict workflow.
3. Updated `config.py` with structured logging, validation schema, and model registry constants.
4. Updated `model_training.py` with tuned hyperparameters for improved models.
5. Updated `data_processing.py` to use structured logging.
6. Created integration tests (`test_integration.py`) — 5 tests covering full pipeline flow.
7. Updated unit tests (`test_pipeline.py`) — 6 new validation tests added.
8. All 17 tests pass.
9. Executed notebook with comparison charts and cross-track documentation.
10. Created LaTeX reports (this summary and the main pipeline integration report).

3. What I Improved from Week 5

- Validation checks prevent incorrect data from propagating through the pipeline.
- Structured logging enables execution traceability (configurable via `LOG_LEVEL`).
- The `HealthConnectPipeline` class replaces manual multi-step orchestration with a single callable object.
- Improved model hyperparameters (e.g., Gradient Boosting recall improved +0.015).
- Integration tests verify that preprocessing and model components interact correctly.

4. What I Integrated

The Data Science candidate model configuration and error analysis were integrated into `get_improved_models()`. The prediction output schema matches the Week 4 architecture specification (`appointment_id`, `probability`, `risk_tier`, `recommended action`).

5. Which Track(s) I Collaborated With

Data Science track. The collaboration involved receiving the candidate model and error analysis (findings: false negatives are the primary concern; `booking_lead_days` dominates feature importance), and providing technical constraints on model interface compatibility and preprocessing format requirements.

6. What Was Exchanged

Data Science provided tuned Random Forest parameters and error analysis findings. ML Engineering provided the prediction output schema and preprocessing format requirements that the Data Science model must conform to.

7. What Changed as a Result

Week 6 improved models use tuned hyperparameters informed by DS error analysis. The pipeline validates prediction output against the agreed schema. Logging provides execution traceability for debugging and audit.

8. Key Findings or Development Outcomes

- Gradient Boosting recall improved from 0.596 to 0.611 under tuned configuration.
- Logistic Regression remains the best model (recall 0.628 CV, 0.647 held-out).
- All 17 tests pass, confirming pipeline correctness.
- The integrated pipeline produces valid risk-stratified predictions.

9. Major Challenges

- Tuning LightGBM while maintaining balanced class weights caused a slight recall drop; requires further investigation.
- Ensuring prediction output schema compatibility between Data Science and ML Engineering required explicit interface documentation.

10. Important Decisions

- Logistic Regression retained as the primary reporting baseline for interpretability.
- `GroupKFold` by `patient_id` deliberately deferred to Week 7 to avoid scope creep in the integration week.
- Validation checks raise exceptions (fail-fast) rather than warnings to prevent silent pipeline corruption.

11. Remaining Issues

- GroupKFold by `patient_id` is still needed for rigorous leakage control.
- Model artefact loading from disk needs end-to-end validation.
- LightGBM tuned configuration requires further hyperparameter search.
- Drift monitoring not yet implemented.

12. Contribution to the Overall HealthConnect Project

The ML Engineering track now provides a validated, integrated pipeline that can receive a model from the Data Science track, validate inputs and outputs, generate risk-stratified predictions, and persist artefacts to disk. This is the foundation for the FastAPI inference service and production deployment planned for Weeks 7–8.

13. Proposed Focus for Week 7

1. End-to-end validation with GroupKFold grouped by `patient_id`.
2. Threshold tuning for F_2 -score optimisation.
3. Model artefact loading and prediction from saved files.
4. FastAPI inference service integration.
5. MLflow experiment tracking and Evidently AI drift monitoring.